

during bent-knee sit-ups (Juker, McGill, Kropf, and Steffen, 1998), not lower as has been previously thought. This is because the hip flexion torque must come from somewhere, and the shortened psoas must contract to higher levels of activation given its compromised length.

Given that the sit-up imposes such a large compression load on the spine, regardless of whether the leg is bent or straight, the issue is not which type of sit-up should be recommended. Rather, sit-ups should not be performed at all by most people. Far better ways exist to preserve the abdominal muscle challenge while imposing lower spine loads. Those who are training for health never need to perform a sit-up; those training for performance may get better results by judiciously incorporating them into their routine.

Part III of this book (Low Back Rehabilitation) offers specific preferred exercises and challenges to specific muscles, but a few flexion exercises are reviewed here. First, hanging with the arms from an overhead bar and flexing the hips to raise the legs is often thought to impose low spine loads because the body is hanging in tension—not compression. This is faulty logic. This hanging exercise generates well over 100 Nm of abdominal torque (Axler and McGill, 1997). This produces almost maximal abdominal activation, which in turn imposes compressive forces on the spine (see [table 4.7](#)). (Note: Hanging with bent knees resulted in higher average spine loads because the imprecise technique employed by many subjects creates substantial lumbar flexion. Further, few subjects were able to maintain form during the straight-leg hang. Good form is important but requires substantial athletic ability.) Similar activation levels can be achieved with the side bridge (shown later and discussed in detail) with lower spine loads.

This having been stated, those who are not interested in sparing their backs and are training with performance objectives may benefit from the high psoas challenge, together with rectus abdominis and oblique activity. Clearly, the curl-up primarily targets the rectus (both upper and lower), and generally, other exercises should be performed to train the obliques. Some have suggested a twisting curl-up to engage the obliques, but this results in a poor ratio of oblique muscle challenge to spine compression compared to the side bridge exercise (Axler and McGill, 1997)—making the side bridge a preferred exercise ([table 4.7](#)).

Table 4.7 Low Back Moment, Abdominal Muscle Activity, and Lumbar Compressive Load During Abdominal Exercises

	MUSCLE ACTIVATION			
	Moment (Nm)	Rectus abdominis (%MVC)*	External oblique	Compression (N)
Straight-leg sit-up	148	121	70	3,506
Bent-leg sit-up	154	103	70	3,350
Curl-up, feet anchored	92	87	45	2,009
Curl-up, feet free	81	67	38	1,991
Quarter sit-up	114	78	42	2,392
Straight-leg raise	102	57	35	2,525
Bent-leg raise	82	35	24	1,767
Cross-knee curl-up	112	89	67	2,964
Hanging, straight leg	107	112	90	2,805
Hanging, bent leg	84	78	64	3,313
Isometric side bridge	72	48	50	2,585

*MVC contractions were isometric. Activation values higher than 100% are often seen during dynamic exercise.

Practical Application: The Paradox of the Sit-Up as a Fitness Test

Schools and the military have long incorporated the sit-up as part of annual fitness testing. Curiously, one branch of the U.S. government (NIOSH) is warning against compressive loads exceeding 3,400 N because they document an increased risk of back injury. Yet another branch, the military services, is incorporating sit-ups as part of its annual fitness tests (recall that they exceed the NIOSH limit). Recognizing this paradox, coupled with the incidence of back injury, branches of the U.S. military have assigned task forces to come up with an alternative. Because measuring components of core fitness is worthwhile, the U.S. Navy, for example, is changing its standards to replace the sit-up with the plank exercise (Peterson, 2013). (They quote many of the norms we have published from various populations over the years: McGill et al., 2013a, 2013b, 2012, 2010).

Loads on the Low Back During Pushing and Pulling

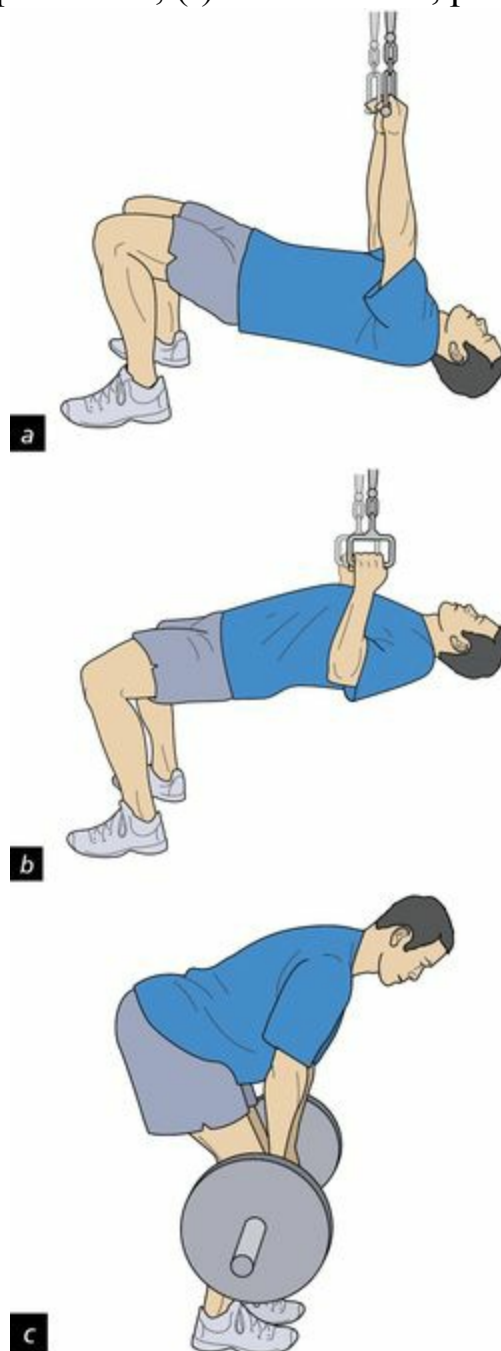
Our insights into the mechanics of pushing and pulling were obtained from our investigations originally intended to set safe occupational load limits (see Lett and McGill, 2006). Both novice (senior university students) and expert (firefighters) pushers/pullers were recruited. The expert

firefighters experienced much lower spine forces. They cleverly used body mechanics with body lean to generate the driving force and to direct the hand forces through the low back, resulting in very small low back moments. The implication is that, once again, the motion and motor patterns that the firefighters elected to use resulted in their superior performance and safety. In this way the magnitude of the push/pull loads became much less important for the back. Technique was dominant!

The technique used in pushing and pulling is very dominant in determining the load on the back. The magnitude of the hand forces is almost irrelevant until extremely high pushing and pulling forces are required. Specifically, hand forces are skillfully directed through the low back. In this way they do not create a moment that would require extra muscle force to counterbalance. For athletes such as rugby players and strongman competitors pulling a bus, this same technique is used. Here the pushing or pulling force is directed through the lumbar spine. Skill is developed to enhance the foot grip, thereby ensuring minimal joint torque.

Pulling exercises generally engage the whole-body linkage with focused challenge to the posterior chain. For example, when standing and performing three variations of rowing exercises, the bent-over row focuses on the low back; the upright cable pull, on the upper back; and the standing one-arm cable pull, on the core stabilizers (Fenwick et al., 2009) (see [figures 4.12](#) and [4.13](#)). The use of suspension straps adds lability to pulling, which generally increases the amount of muscle activation (see [figure 4.14](#) and [table 4.8](#)). Because tasks with suspension straps are more challenging, exercise choice would be guided by the person's injury history, training goals, and current fitness level and tolerance.

Figure 4.12 Three variations of pulling create different changes. The inverted row focuses on the upper back, the traditional row focuses on the low back, and the standing one-arm cable pull primarily challenges the antitwist mechanism of the core: (a) TRX pull up, position 1; (b) TRX pull up, position 2; (c) bentover row, position 1; (d) bentover row, position 2; (e) one-arm row, position 1; (f) one-arm row, position 2.



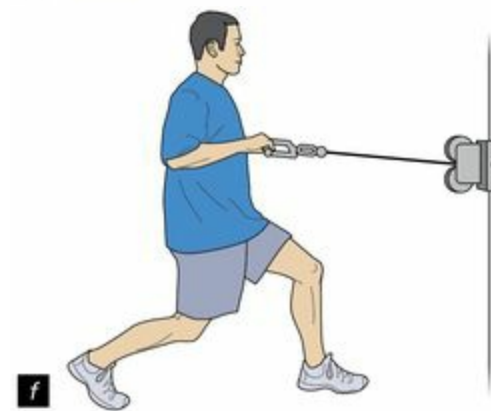


Figure 4.13 Average muscle activation of the figure 4.12 exercises.

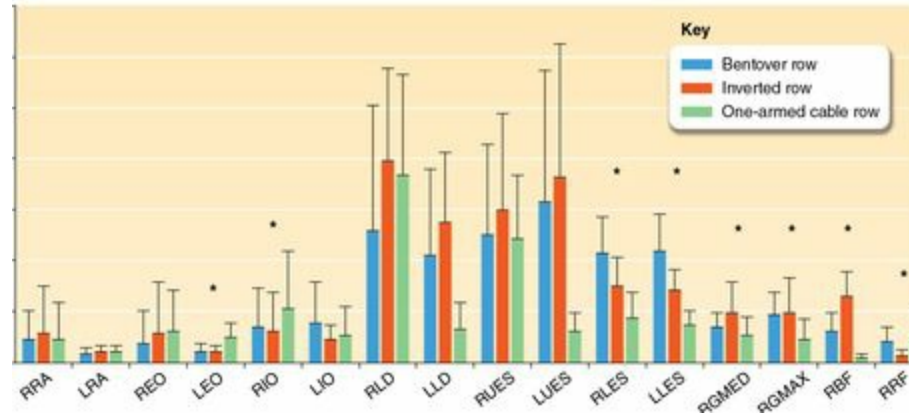


Figure 4.14 TRX suspension straps create a range of pulling and pushing exercise challenges simply by changing the angle of the body. Spine-conserving technique requires stiffening the core for pain control, centering the scapulae (i.e., the first part of the exercise is to depress the scapulae and activate the latissimus dorsi), and using a robust handgrip to radiate force up the arm linkage.

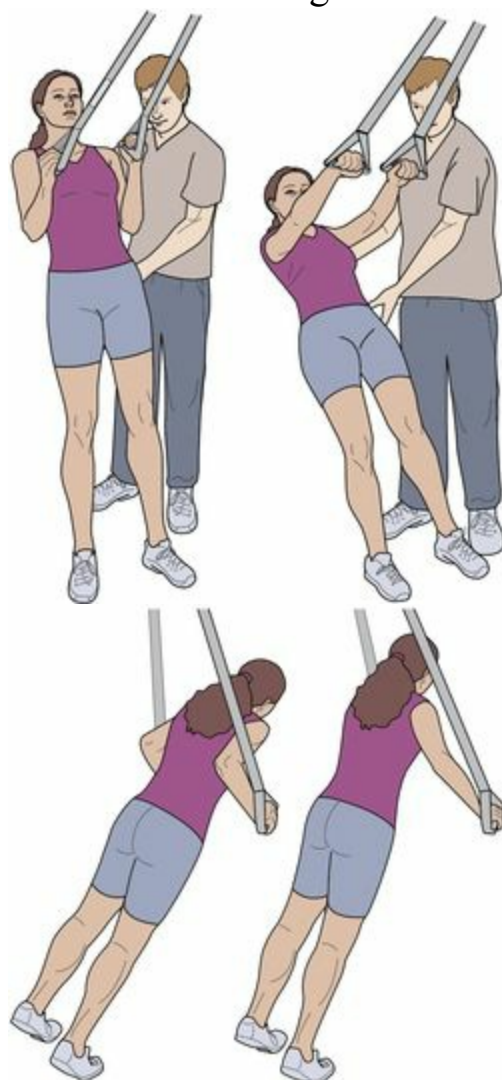


Table 4.8 Pulling Exercises Ranked on Compression Load Imposed on the Low Back

Exercise	Rank	Mean spine compression (N)	Standard deviation
Pull-up	1	2,852.3	1,339.1
Chin-up	2	2,679.8	1,327.0
TRX pull-up	3	2,626.6	1,181.5
Inverted row using body weight	4	2,294.3	767.0
TRX pull—angle 2	5	2,287.6	764.2
Stable shoulder retraction, coached	6	2,221.3	774.9
TRX pull—angle 1	7	2,041.8	715.1
Stable shoulder retraction, not coached	8	1,997.1	633.6
TRX shoulder retraction, coached	9	1,706.7	620.3

Rank of mean spine compression during the peak pulling phase of each pulling exercise. The TRX exercises use suspension straps adding lability to the exercise. For a complete description of each exercise, as well as a description of the muscle activation level, refer to McGill, Cambridge, and Andersen, 2014).

Loads on the Low Back During Pushing Exercises

Many have recognized that many forms of spine stabilization exercise engage the abdominal hoop comprising rectus abdominis, the internal and external obliques, and transverse abdominis in an isometric contraction (McGill, 2006). For this reason, push-up exercises are sometimes used as torso training exercises. Clinical observation confirms that performing push-ups elicits back pain in some patients, yet others find push-ups relieving. In our quantification of push-up exercises, we examined styles ranging from traditional to placing the hands on labile surfaces (balls), staggered hand placement, one arm push-ups, and so on (Freeman et al., 2006). Although performing push-ups with the hands on a labile surface such as a ball has some effect on spine load, the one-arm and more ballistic forms of the exercise that require the hands to move are much more demanding on the spine (see [table 4.9](#)). Our recent work on performing pushes with suspension straps illustrates how muscle challenge and muscle demand can be influenced by this lability (McGill, Cannon, and Andersen, 2014a) and adjusted with the angle of the body (McGill et al., 2014b) (see [figure 4.15](#) and [table 4.10](#)). On average, the hanging straight-leg raise created approximately 3,000 N of spine compression, whereas the body saw created less than 2,500 N. The hanging straight-leg raise created the highest challenge to the abdominal wall (>130% of MVC in rectus abdominis, 88% of MVC in external oblique). The body saw resulted in almost 140% of MVC activation of the serratus anterior. Those interested in challenging the abdominal obliques and steering the asymmetrical force from staggered hand placement through the torso will be

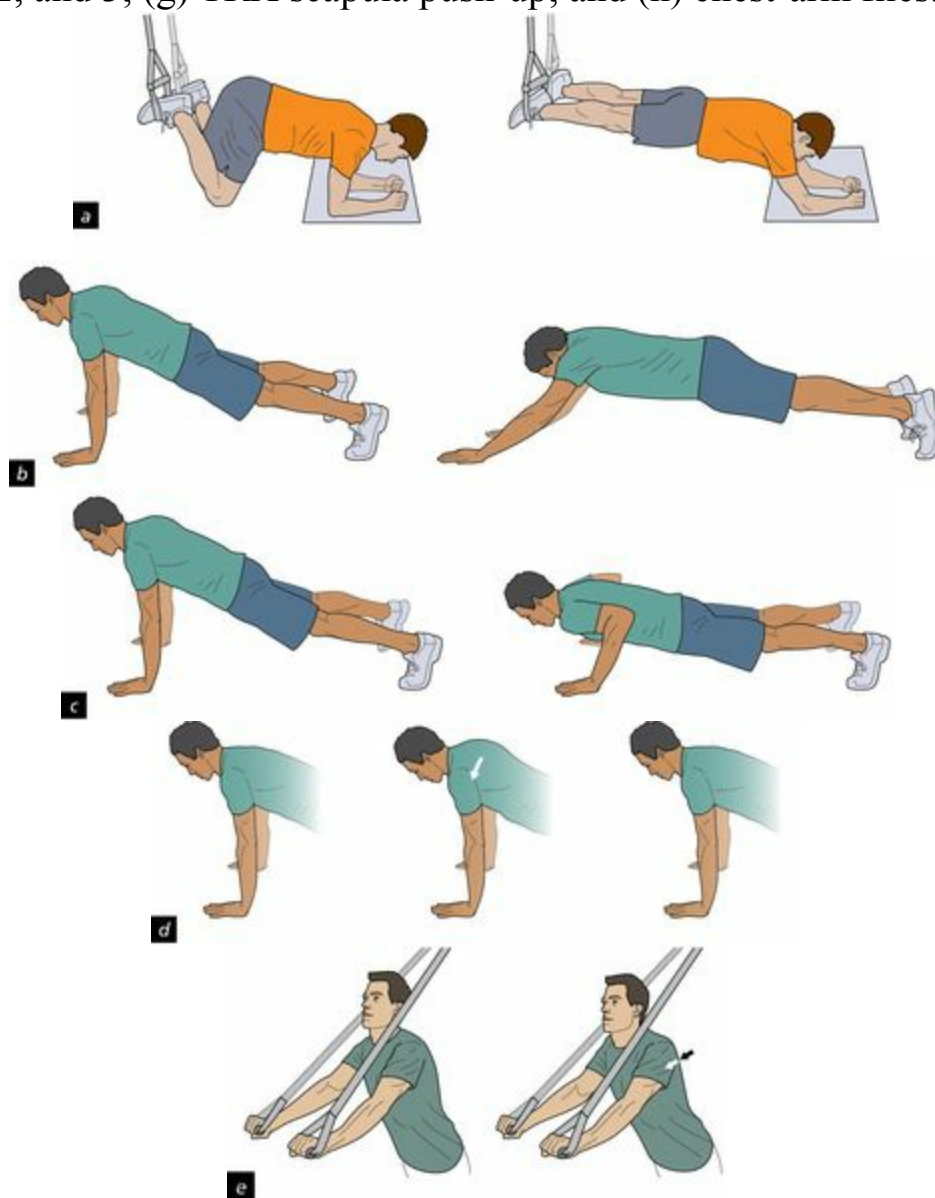
interested in the quite modest increase in spine compression demand. Not surprisingly, the plyometric forms of the push-ups are much more muscularly demanding and therefore result in higher spine load. This may be a concern for those who are sensitive to spine compression during provocative diagnostic testing. Spinal loading during many forms of the push-up is substantial. There is little wonder that these exercises are problematic for some people with painful backs. On the other hand, they may be very appropriate as an abdominal plyometric exercise for high-performance athletes.

Table 4.9 Total Compression While Performing Various Styles of Push-Ups

Exercise	Compression (N)
Standard	1,838
One-arm	5,848
Staggered hands (right forward)	2,532
Staggered hands (left forward)	2,337
Right hand on ball	2,315
Left hand on ball	2,295
Two hands on one ball	2,840
Hands on two balls (one on each)	2,829
Alternating	6,224
Clapping	4,699
Fast concentric	3,905
Slow eccentric	2,222

Adapted, by permission, from S. Freeman, A. Karpowicz, J. Gray, and S.M. McGill, 2006, "Quantifying muscle patterns and spine load during various forms of the pushup." *Medicine and Science in Sports and Exercise*, 38(3): 570-577.

Figure 4.15 (a) Body saw; (b) walkout; (c) standard push-up; (d) stable shoulder protraction; (e) TRX shoulder protraction; (f) TRX pushes at angles 1, 2, and 3; (g) TRX scapula push-up; and (h) chest-arm flies.



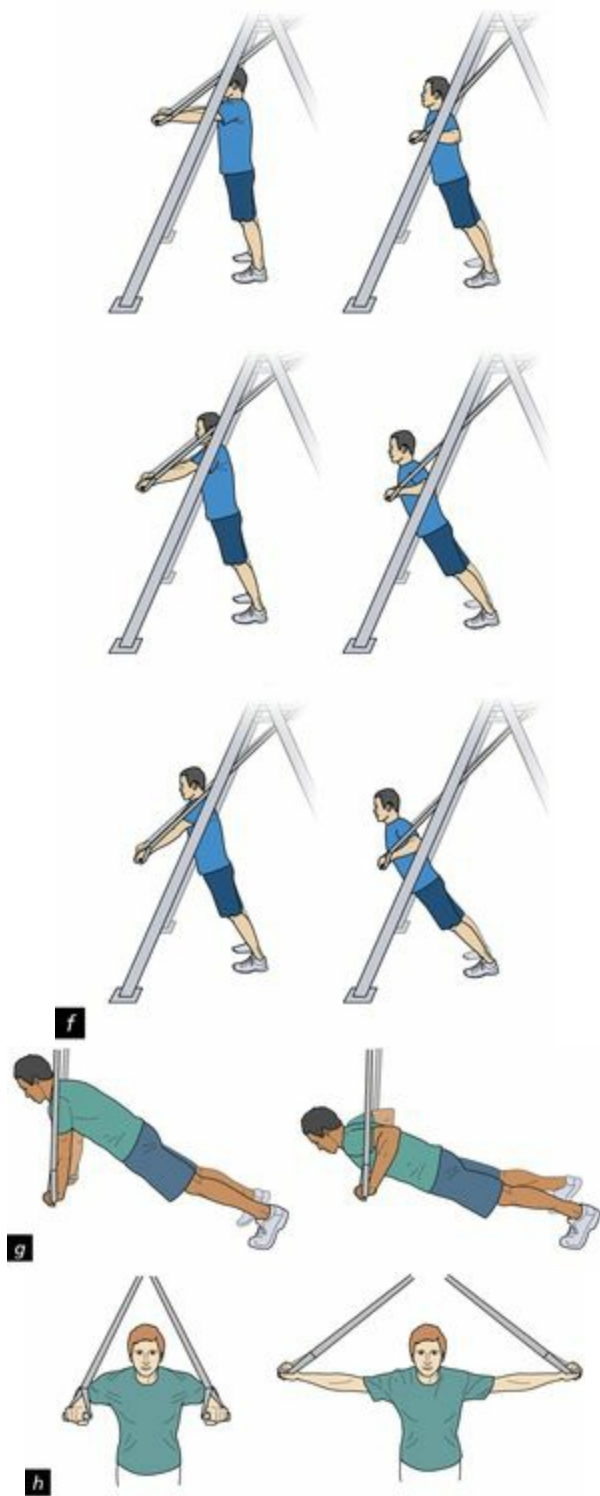


Table 4.10 Push Exercises Ranked on Compression Load Imposed on the Low Back

Exercise	Rank	Mean spine compression N)	Standard deviation
TRX push—angle 3	1	1,838.9	852.9
TRX push-up	2	1,653.4	759.8
TRX push—angle 2	3	1,631.1	712.0
Stable shoulder protraction, coached	4	1,582.7	717.3
TRX scapula push-up	5	1,581.0	505.6
TRX shoulder protraction, coached	6	1,528.2	707.7
TRX push—angle 1	7	1,484.9	638.3
TRX shoulder protraction, not coached	8	1,449.5	566.8
Standard push-up	9	1,399.2	716.7
Stable shoulder protraction, not coached	10	1,381.0	515.3

Rank of mean spine compression at the peak force phase of each push exercise. We also tested whether coaching shoulder centration patterns during pushing reduced joint load.

Loads on the Low Back During Extension Exercises

As with the flexion exercises discussed previously, plenty of EMG-based studies have explored extension exercises, but only one attempted to quantify the resulting tissue loads. Exercise prescriptions will not be successful if the spine loading is not constrained for those with bad backs. Using the virtual spine approach, Callaghan, Gunning, and McGill (1998) attempted to rank extension exercises on the muscle challenge, the resultant spine load, and their optimal ratio. The key to preserving a therapeutic muscle activation level while minimizing the spine load is to activate only one side of the spine musculature at a time. The muscle anatomy section in chapter 3 describes the functional separation of the thoracic and lumbar portions of the longissimus and iliocostalis. For the purposes of this discussion, we can think of the extensors in four sections—right and left thoracic portions and right and left lumbar portions. The common extension task of performing torso extension with the legs braced and the cantilevered upper body extending over the end of a bench or Roman chair ([figure 4.16a](#)) activates all four extensor groups and typically imposes over 4,000 N of compression on the spine. Even worse is the commonly prescribed back extension task in clinics, in which the patient lies prone and extends the legs and outstretched arms; this again activates all four extensor sections but imposes up to 6,000 N on a hyperextended spine ([figure 4.16b](#)). This is not justifiable for any patient!

Figure 4.16 Specific extension exercises quantified for muscle activation and the resultant spine load (shown in table 4.11): (a) trunk extension, (b) prone leg and trunk extension, (c) single-leg extension, and (d) single-leg and contralateral arm extension (bird dog).

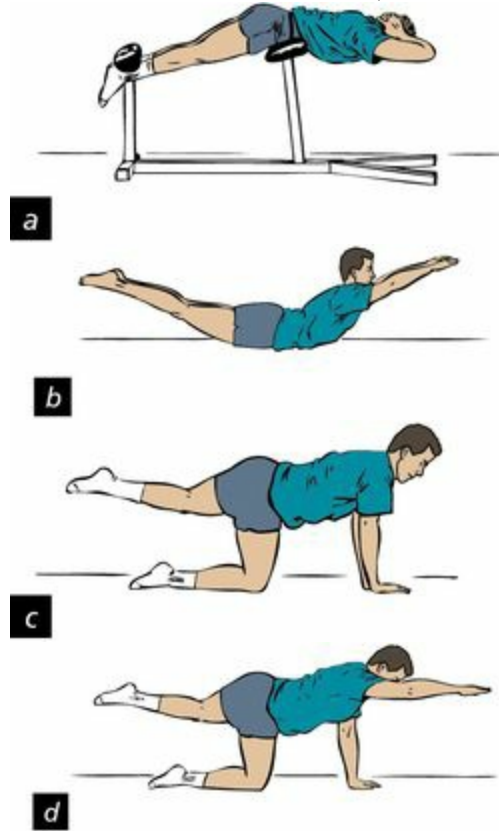


Table 4.11 Mean Activation Levels (± 1 SD) of 14 EMG Channels for 13 Subjects Performing a Variety of Extensor-Dominant Exercises (Expressed as %MVC)

Electromyographic channel*	EXTENSION						Calibration posture
	Right leg	Left leg	Right leg and left arm	Left leg and right arm	Trunk and legs	Trunk	
Right RA X	3.3	2.7	4.0	3.5	4.7	3.1	1.4
SD	2.4	1.9	2.0	2.0	2.2	1.8	1.0
Right EO X	8.4	4.9	16.2	5.2	4.3	3.7	1.0
SD	4.9	1.5	6.0	2.3	2.5	1.7	0.6
Right IO X	12.0	8.2	15.6	12.0	12.1	12.7	1.9
SD	6.8	2.5	8.2	4.2	10.1	10.8	1.2
Right LD X	8.1	5.8	12.0	12.5	11.2	6.5	5.9
SD	5.4	3.5	9.6	6.2	4.3	4.0	8.5
Right TES X	5.7	13.7	11.5	46.8	66.1	45.4	21.0
SD	2.0	7.5	6.6	29.3	18.8	10.6	9.0
Right LES X	19.7	11.7	28.4	19.4	59.2	57.8	21.3
SD	9.1	4.9	10.2	11.0	11.7	8.5	4.6
Right MF X	21.9	10.8	31.5	16.1	51.9	47.5	16.4
SD	6.3	6.0	8.2	12.0	14.7	12.3	5.6
Left RA X	4.3	3.6	4.4	4.2	6.5	3.7	2.2
SD	3.4	3.6	3.8	3.9	3.4	2.4	2.1
Left EO X	5.4	9.0	6.2	15.9	6.3	5.2	1.8
SD	2.0	3.8	2.5	6.6	3.2	5.2	1.0
Left IO X	16.0	11.3	22.6	15.2	11.0	12.5	1.6
SD	8.6	7.0	9.2	6.7	5.9	6.1	1.3
Left LD X	4.5	5.0	10.7	6.2	9.2	5.1	6.1
SD	4.3	4.5	18.2	4.4	5.1	4.1	8.5
Left TES X	15.0	4.5	42.9	10.5	63.6	41.6	21.2
SD	7.5	2.0	20.5	5.9	22.7	10.0	9.8
Left LES X	11.3	16.8	19.5	25.5	56.8	57.0	23.3
SD	6.6	4.5	7.4	7.3	14.5	14.7	8.4
Left MF X	11.9	22.3	16.6	33.8	57.3	53.3	18.7
SD	7.0	6.1	7.2	6.7	11.4	12.0	4.3

*Electromyographic channel: RA = rectus abdominis muscle; EO = external oblique muscle ; IO = internal oblique muscle ; LD = latissimus dorsi muscle; TES = thoracic erector spinae muscle; LES = lumbar erector spinae muscle; MF = multifidus muscle. Calibration posture: standing, trunk flexed to 60°, neutral lumbar posture, 10 kg (22 lb) held in hands with arms hanging straight down.

Several variations of exercise technique can preserve activation in portions of the extensors and greatly spare the spine of high load. For example, kneeling on all fours and extending one leg at the hip generally activates one side of the lumbar extensors to over 20% of maximum and imposes only 2,000 N of compression ([figure 4.16c](#)). Performing the bird dog, in which the opposite arm is extended at the shoulder while the leg is raised ([figure 4.16d](#)), adds activity to one side of the thoracic extensors (generally around 30 to 40% of maximum) and contains the spine load to about 3,000 N. In addition, the special techniques shown for this exercise in chapter 10 attempt to enhance the motor control system to groove stabilizing patterns. For data describing these exercises, see [table 4.11](#).

Loads on the Low Back During Gait: Walking, Carrying, and Using

Elliptical Trainers

Walking has already been discussed and forms a basic pattern for the ambulation tasks. When carrying a handheld load while walking, what is the most spine-friendly tactic? Carrying loads in one hand causes more spine load than doubling the load by carrying the same load in each hand. For example, when carrying 30 kg (66 lb) in one hand, the low back compression exceeded 2,800 N; however, splitting the load between hands reduced low back compression to 1,570 N (a reduction of 44%). Doubling the total load by carrying 30 kg in each hand (total of 60 kg) actually produced lower spine compression than carrying 30 kg in one hand. This is due to the additional muscle activity required to support the lateral bending moment with uneven frontal plane loading (McGill, Marshall, and Andersen, 2013). The message for those with back pain is to split the load between the hands ([figure 4.17](#)).

Other carrying exercises such as the kettlebell carry, known as bottoms-up, change the stability requirements which the motor control scheme addresses by using more bilateral torso muscle coactivation ([figure 4.18](#)). Such activities are modest and conservative in terms of influencing spine load (McGill and Marshall, 2012).

Figure 4.17 Carrying loads in one hand or splitting the load between two hands greatly unloads the spine. This effect is greater with heavier loads as they dominate the effect of body mass.

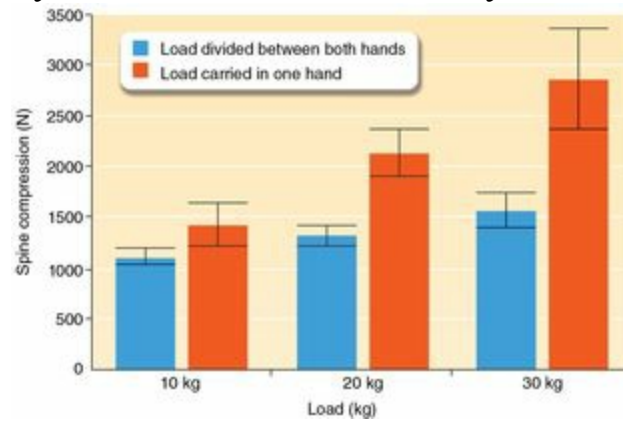


Figure 4.18 The kettlebell bottoms-up carry enhances the control requirements while restricting the amount of load that can be carried and the torso muscle activity.



Walking and exercising on an elliptical trainer require quite different body linkage mechanics. Our study was motivated by the observation that some people reported back pain relief with elliptical trainer exercise, whereas others reported that their pain was exacerbated. Those with tight hips and normal hip range of motion were recruited. First, the elliptical trainer requires more hip motion than normal walking (Moreside and McGill, 2012). On the other hand, elliptical trainers result in much higher gluteal muscle activation, which may be a training target for some people. Those with tight hips endured more spine motion and a more forward leaning torso, increasing the spine load. Thus, we recommend hip screening and pain provocation tests, as well as a pain history, before suggesting elliptical training.

Loads on the Low Back During Standing Exercises

Many exercises, and combinations of exercises, are performed in a standing posture. Given the variety of loads and technique, I refer you to our

original studies:

- Standing torso exercises: McGill, S.M., Karpowicz, A., and Fenwick, C. (2009b) Exercises for the torso performed in a standing posture: Motion and motor patterns. *Journal of Strength and Conditioning Research* 23 (2): 455-464.
- Standing body blade exercises: Moreside, J.M., Vera-Garcia, F.J., and McGill, S.M. (2007) Trunk muscle activation patterns, lumbar compressive forces and spine stability when using the body blade. *Physical Therapy*, 87 (2): 153-163.
- Ski simulators: Banerjee, P., Brown S., Howarth, S., and McGill, S.M. (2009) Torso and hip muscle activity and resulting spine load and stability while using the Profitter 3-D Cross Trainer. *Journal of Applied Biomechanics*, 25: 73-84.
- Hula hooping: McGill, S.M., Cannon, J., and Andersen, J. (Submitted) Physiological and biomechanical mechanisms in hula hooping: Caloric expenditure and spine loads.

Loads on the Low Back During Coitus

Every primary care clinician reports being often asked by couples with back pain about less-painful or pain-free sexual positions. Until now, clinicians had no scientific foundation to guide them. We appear to have been the first group to document spine and hip motion, and muscle activation levels, during coitus. Given the variety of positions and intensity of effort, we simply direct you to our recent medical publications to appreciate the spectrum of loads: Sidorkewicz and McGill (2014a and 2014b) and McGill (2015).

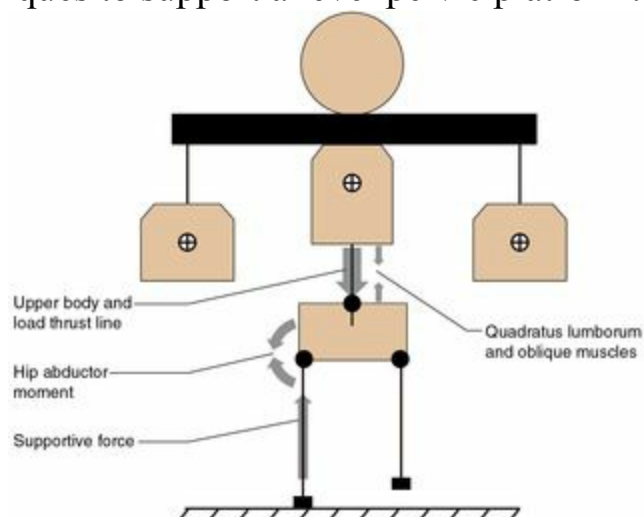
Loads on the Low Back During More Demanding Forms of Exercise

Kettlebell swings have gained popularity; some athletes claim that the exercise was vital in their rehabilitation following disc injury. Yet other very athletic people claim that swings are the one exercise that instigates pain. This motivated our investigation, in which we found that kettlebell swings create a desirable hip-hinge squat pattern characterized by rapid muscle activation and relaxation cycles of substantial magnitude (50% of MVC for

the low back extensors and 80% of MVC for the gluteal muscles with a 16 kg [35 lb] kettlebell), which results in about 3,200 N of low back compression. Some unique loading patterns discovered during the kettlebell swing included the posterior shear of the L4 vertebra on L5, which is opposite in polarity to a traditional lift. Thus, the quantitative analysis provided an insight into why many people credit kettlebell swings with restoring and enhancing back health and function, although a few find that they irritate tissues. It appears that those with sufficient shear stability of the lumbar joints may do well with swings, whereas those deficit in shear stability should perhaps perform the shear instability tests described in this book.

In one study we measured competitive strongman athletes as they performed events such as the farmer's walk, super yoke ([figure 4.19](#)), Atlas stone lift, suitcase carry, keg walk, tire flip, and log lift (McGill, McDermott, and Fenwick, 2009). The results document the unique demands of these whole-body events and, in particular, the demands on the back and torso. For example, the very large moments required at the stance-leg hip for abduction when performing a yoke walk exceed the strength capability of the hip. Here, muscles such as quadratus lumborum made up for the strength deficit by generating frontal plane torque to support the torso and pelvis. In this way, the stiffened torso acts as a source of strength to allow joints with insufficient strength to be buttressed, resulting in successful performance. We are often asked why we study world-class performance. Here we see a wonderful example of obtaining insight that explains many lower-level rehabilitation challenges. Children with a paralyzed quadratus lumborum muscle have extreme difficulty walking. Most gait biomechanists have concluded that frontal plane support of the pelvis is supplied by the hip abductor muscles. The insight obtained here demonstrated that the opposite core muscles must function with the gluteal muscle to create walking proficiency.

Figure 4.19 Measuring world-class strongmen revealed how they carried loads that exceeded their strength. We measured a hip abduction strength of approximately 500 Nm, yet 750 Nm of strength was required on the support hip to complete the super yoke carry. The missing strength appears to have been supplied by the contralateral quadratus lumborum and abdominal obliques to support a level pelvic platform.



Future Trends for Linking Back Load and Exercise for People With Osteoporosis

Matching exercise with the specific load-bearing tolerance of a person is critical for fragile osteoporotic spines and otherwise injured backs. However, metrics such as vertebral bone density (measured with DEXA, for example) have not proven to be very accurate in representing compressive strength. Advances by Professor Keaveney's group (e.g., Wang, Videman, and Battié, 2012) have resulted in a CT-based bone quality index that incorporates density with the connectivity of the trabecular bone—resulting in a better prediction of strength. Our work has identified spine loads with various activities. Here we put the two approaches together to create a rough guide for selecting activity based on bone quality and load-bearing ability (see [table 4.12](#)). For example, a person with a bone quality index of 0.7 can walk safely without risk of further fracture. Having a quality index of 0.8 would allow the person to add one-legged quadruped exercises and retain a margin of safety.

Table 4.12 Ranking of Activities Based on Estimates of Spine Load and Keaveney's Bone Quality Index

Minimum	aBMD (g/cm ³)	Activity
0.7		Walking (swing arms)
0.8	Add	Curl-up, one-leg quadruped
0.9		
1.0	Add	Push-up
1.1	Add	Sit-up, kettlebell swing with <16 kg (35 lb)
1.2	Add	Superman, suspended push-up
1.3	Add	Rowing (in boat)
1.4		
1.6	Add	Golf, one-arm push-up, box lift 50 lb (22.7 kg) (knee to waist)

This index is linked with our many studies quantifying joint load to create this ranked listing of activities that, on average, should be tolerable. We have incorporated a safety factor of a spine load that is twice the tolerance load.

For those with quantifiable osteoporosis, we can use their bone quality score to match tolerable exercise and activity. Keaveney's work can guide this exercise selection while also building in a predetermined margin of safety. We chose half of the tolerable load as a reasonable compromise between promoting a variety of activities and safety.

Dubious Lifting Mechanisms

In the 1950s and 1960s, spine biomechanists faced a paradox. The simple spine models of the day predicted that the spine would be crushed to the point of injury during certain lifting tasks, yet when people performed those tasks, they walked away uninjured. This motivated several research groups to theorize about mechanisms that could unload compressive stresses from the spine. Researchers proposed three major mechanisms: the intra-abdominal pressure mechanism, the lumbodorsal fascia mechanism, and the hydraulic amplifier. Although none has survived scrutiny, clinical belief in these mechanisms remain. Nonetheless, some components of these mechanisms provided insight for subsequent studies and led to the understanding that we have today. For this reason they are reviewed briefly here.

Intra-Abdominal Pressure

Does intra-abdominal pressure (IAP) play an important role in the support of the lumbar spine, especially during strenuous lifting, as has been claimed for many years? Anatomical accuracy in representation of the involved tissues has been influential in this debate. Further, research on lifting mechanics has formed a cornerstone for the prescription of abdominal belts for industrial workers and has motivated the prescription of abdominal strengthening programs. Many researchers have advocated the use of IAP as a mechanism to directly reduce lumbar spine compression (Bearn, 1961; Thomson, 1988). However, some researchers believe that the role of IAP in reducing spinal loads has been overemphasized (Grew, 1980; Krag et al., 1986).

Anatomical Consistency in Examining the Role of Intra-Abdominal Pressure

Morris, Lucas, and Bresler (1961) first operationalized the mechanics of the original proposal into a model and described it as follows. Pressurizing the abdomen by closing the glottis and bearing down during an exertion exerts hydraulic force down on the pelvic floor and up on the diaphragm,

creating a tensile effect over the lumbar spine or at least alleviating some of the compression. Missing in the early calculations of this hydraulic potential was the full acknowledgment of the necessary abdominal activity (contracting the abdominal wall imposes extra compression on the spine). But an evaluation of the trade-off between the extra abdominal muscle compression and the hydraulic relief depended on the geometrical assumptions made. Some of these assumptions appear to be outside of biological reality. In fact, experimental evidence suggests that somehow, in the process of building up IAP, the net compressive load on the spine is increased! Krag and coworkers (1986) observed increased low back EMG activity with increased IAP during voluntary Valsalva maneuvers. Nachemson and Morris (1964) and Nachemson, Andersson, and Schultz (1986) showed an increase in intradiscal pressure during a Valsalva maneuver, indicating a net increase in spine compression with an increase in IAP, presumably a result of abdominal wall musculature activity.

In our own investigation, in which we used our virtual spine model, we noted that net spine compression was increased from the necessary concomitant abdominal activity to increase IAP. Furthermore, the size of the cross-sectional area of the diaphragm and the moment arm used to estimate force and moment at the lower lumbar levels, produced by IAP, have a major effect on conclusions reached about the role of IAP (McGill and Norman, 1987). The diaphragm surface area was taken as 243 cm^2 , and the centroid of this area was placed 3.8 cm anterior to the center of the T12 disc (compare these values with those used in other studies: 511 cm^2 for the pelvic floor, 465 cm^2 for the diaphragm, and moment arm distances of up to 11.4 cm, which is outside the chest in most people). During squat lifts, the net effect of the involvement of the abdominal musculature and IAP seems to be to increase compression rather than alleviate joint load. (A detailed description and analysis of the forces are in McGill and Norman, 1987.) This theoretical finding agrees with the experimental evidence of Krag and colleagues (1986), who used EMG to evaluate the effect of reducing the need for the extensors to contract (they didn't), and of Nachemson, Andersson, and Schultz (1986), who documented increased intradiscal pressure with an increase in IAP.

Role of IAP During Lifting

The generation of appreciable IAP during load-handling tasks is well documented. The role of IAP is not. Farfan (1973) suggested that IAP creates a pressurized visceral cavity to maintain the hooplike geometry of the abdominal muscles. In recent work in which they measured the distance of the abdominal muscles to the spine (moment arms), McGill, Juker, and Axler (1996) were unable to confirm substantial changes in geometry when the abdominal muscles were activated in a standing posture. However, the compression penalty of abdominal activity cannot be discounted. The spine appears to be well suited to sustain increased compression loads if intrinsic stability is increased. An unstable spine buckles under extremely low compressive loads (e.g., approximately 20 N) (Lucas and Bresler, 1961). The geometry of the spinal musculature suggests that individual components exert lateral and anteroposterior forces on the spine that can be thought of as guy wires on a mast to prevent bending and compressive buckling (Cholewicki and McGill, 1996). Moreover, activated abdominal muscles create a rigid cylinder of the trunk, resulting in a stiffer structure. Both Cholewicki and colleagues (1999) and Grenier and McGill (2007, 2008) documented increased torso stiffness during elevated IAP even when accounting for similar abdominal wall contraction levels. It also appears that IAP can influence pelvic mechanics and pain. Mens and colleagues (2006) noted higher pelvic ring forces with elevated IAP, which may stabilize some yet destabilize others—depending on the nature of tissue compromise. The clinical solution for this divergence is to perform provocative testing to reveal whether the patient's problem is helped or exacerbated. Thus, although the increased IAP commonly observed during lifting and in people experiencing back pain does not have a direct role in reducing spinal compression or in adding to the extensor moment, it does stiffen the trunk and prevent tissue strain or failure from buckling.

Lumbodorsal Fascia

Various mechanical roles have been attributed to the lumbodorsal fascia (LDF). In particular, some have suggested that the LDF reduces spine loads—solving the paradox noted earlier. In fact, some have recommended lifting postures based on various interpretations of the mechanics of the LDF. Gracovetsky, Farfan, and Lamy (1981) originally suggested that lateral forces

generated by internal oblique and transverse abdominis are transmitted to the LDF via their attachments to the lateral border. They also claimed that the fascia could support substantial extensor moments. Further, lateral tension from abdominal wall attachments was hypothesized to increase longitudinal tension from Poisson's effect, causing the posterior spinous processes to move together resulting in lumbar extension. This sequence of events formed an attractive proposition because the LDF has the largest moment arm of all the extensor tissues. As a result, any extensor forces within the LDF would impose the smallest compressive penalty to vertebral components of the spine.

Three independent studies, however, examined the mechanical role of the LDF and collectively questioned the idea that the LDF could support substantial extensor moments (Macintosh and Bogduk, 1987; McGill and Norman, 1988; Tesh, Dunn, and Evans, 1987). As previously noted, regardless of the choice of LDF activation strategy, the LDF contribution to the restorative extension moment was negligible compared with the much larger low back reaction moment required to support a load in the hands. Its function may be that of an extensor muscle retinaculum (Bogduk and Macintosh, 1984). Hukins, Aspden, and Hickey (1990) proposed on theoretical grounds that the LDF acts to increase the force per unit of cross-sectional area that muscle can produce by up to 30%. They suggested that this increase in force is achieved by constraining the bulging of the muscles when they shorten. This contention remains to be proven. Tesh, Dunn, and Evans (1987) suggested that the LDF may be important for supporting lateral bending. Furthermore, there is no question that the LDF is involved in enhancing the stability of the lumbar column. No doubt, a complete assessment of these notions will be pursued in the future.

Hydraulic Amplifier

The final mechanism hypothesized to unload compressive stresses from the spine was the hydraulic amplifier. This hybrid mechanism depends on three notions. First, the elevated IAP preserves the hooplike geometry of the abdominal wall during exertion. The IAP must also exert hydraulic pressure posteriorly over the spine and presumably through to the underside of the LDF. Finally, as the extensor muscle mass contracts, it was proposed to bulk

upon shortening, again increasing the hydraulic pressure under the fascia. The biomechanical attraction of the pressure under the fascia is that any longitudinal forces generated in the fascia reduce the need for the underlying muscles to contribute extensor forces, thereby lowering the compressive load on the spine. Both of these proposals were dismissed. Given the size of the fascia, hydraulic pressures would have to reach levels of hundreds of mmHg. Pressures of this magnitude simply are not observed during recording (Carr et al., 1985). Moreover, the presence or absence of IAP makes little difference on the hooplike geometry of the abdominal wall (McGill, Juker, and Axler, 1996), because this is more modulated by the posture.

IAP, LDF, and Hydraulic Amplifier: A Summary

IAP, the mechanical role of the LDF, and the existence of the hydraulic amplifier were proposed to account for the paradox that people were able to perform lifts that the simple models suggested would crush their spines. Yet, although both IAP and the LDF appear to play some role in lifting, none of the three proposed mechanisms was a tenable explanation for the paradox of uncrushed spines under heavy loading, whether considered separately or combined with the other two. The problem lay in the simple models of three and four decades ago. Not only were the rather complex mechanics not represented with the necessary detail, but also the strength of the tissues to bear load was also quite underestimated in the early tests that used old cadaveric samples that were crushed underneath artificially stiff rams of materials-testing machines that caused failure too early. Vestiges of these old notions remain today.

Other Important Mechanisms of Normal Spine Mechanics

Several other features of spine mechanics influence function and ultimately underpin strategies for injury prevention and rehabilitation. The most important are presented in this section.

Biomechanics of Diurnal Spine Changes

Most people have experienced the ease of taking off their socks at night compared to putting them on in the morning. The diurnal variation in spine length (the spine being longer after a night's bed rest), together with the ability to flex forward, has been well documented. Reilly, Tynell, and Troup (1984) measured losses in sitting height over a day of up to 19 mm (0.7 in.). They also noted that approximately 54% of this loss occurred in the first 30 minutes after rising. Over the course of a day, hydrostatic pressures cause a net outflow of fluid from the disc, resulting in narrowing of the space between the vertebrae, which in turn reduces tension in the ligaments. When a person lies down at night, osmotic pressure in the disc nucleus exceed the hydrostatic pressure, causing the disc to expand. Adams, Dolan, and Hutton (1987) noted that the range of lumbar flexion increased by 5 to 6° throughout the day. The increased fluid content after rising from bed caused the lumbar spine to be more resistant to bending, although the musculature did not appear to compensate by restricting the bending range. Adams and colleagues estimated that disc-bending stresses were increased by 300% and ligament stresses by 80% in the morning compared to the evening; they concluded that there is an increased risk of injury to these tissues during bending forward early in the morning. Subsequently, Snook and colleagues (1998) demonstrated that simply avoiding full lumbar flexion in the morning reduced back symptoms. We are beginning to understand the mechanism.

Early-Morning Exercise

Clinical Relevance

People should not undertake spine exercises—particularly those that require full spine flexion or bending—just after rising from bed, given the elevated tissue stresses that result. This would hold true for any occupational task requiring full spine range of motion. Waiting an hour would be wise. Going for a walk would reduce this length of time.

Spinal Memory

The function of the spine is modulated by previous activity. This occurs because the loading history determines disc hydration (and therefore the size of the disc space and disc geometry), which in turn modulates ligament rest length, joint mobility, stiffness, and load distribution. Consider the following scenario: McKenzie (1979) proposed that the nucleus within the annulus migrates anteriorly during spinal extension and posteriorly during flexion. McKenzie's program of passive extension of the lumbar spine (which is currently popular in physical therapy) was based on the supposition that an anterior movement of the nucleus would decrease pressure on the posterior portions of the annulus, which is the most problematic site of herniation. Because of the viscous properties of the nuclear material, such repositioning of the nucleus is not immediate after a postural change, but rather, takes time. Krag and coworkers (1987) observed anterior movement of the nucleus during lumbar extension, albeit quite minute, from an elaborate experiment that placed radio-opaque markers in the nucleus of cadaveric lumbar motion segments. Whether this observation was caused simply by a redistribution of the centroid of the wedge-shaped nuclear cavity moving forward with flexion or was a movement of the whole nucleus remains to be determined. Nonetheless, hydraulic theory would suggest lower bulging forces on the posterior annulus if the nuclear centroid moved anteriorly during extension. If compressive forces were applied to a disc in which the nuclear material was still posterior (as in lifting immediately after a prolonged period of flexion), a concentration of stress would occur on the posterior annulus.

Although this area of research needs more development, a time constant seems to be associated with the redistribution of nuclear material. It appears unwise to lift an object immediately following prolonged flexion, such as sitting or stooping (e.g., a stooped gardener should not stand erect and

immediately lift a heavy object). Furthermore, Adams and Hutton (1988) suggested that prolonged full flexion may cause the posterior ligaments to creep, which may allow damaging flexion postures to go unchecked if lordosis is not controlled during subsequent lifts. In a study of posterior passive tissue creep during sitting in a slouched posture, McGill and Brown (1992) showed that over the 2 minutes following 20 minutes of full flexion, subjects regained only half of their intervertebral joint stiffness. Even after 30 minutes of rest, some residual joint laxity remained. This finding is of particular importance for people whose work is characterized by cyclic bouts of full-end-range-of-motion postures followed by exertion. Before lifting following a stooped posture or after prolonged sitting, a case could be made for standing or even consciously extending the spine for a short period. Allowing the nuclear material to equilibrate, or move anteriorly to a position associated with normal lordosis, may decrease forces on the posterior nucleus in a subsequent lifting task. Ligaments will regain some protective stiffness during a short period of lumbar extension. In conclusion, the anatomy and geometry of the spine are not static. Much research remains to be done to understand the importance of tissue loading history on subsequent biomechanics, rehabilitation therapies, and injury mechanics.

Functional Significance of Spinal Memory

Clinical Relevance

It would appear protective to avoid loading immediately after a bout of prolonged flexion. In the occupational world this has relevance to ambulance drivers, for example, who drive to an accident scene without the luxury of time to warm up (or reset the passive tissues) before lifting. They would be wise to sit with a lumbar pad to avoid lumbar flexion and the associated creep. The athletic world provides good examples as well, such as sitting on the bench before engaging in play. Those with sensitive backs would do well to avoid sitting on the bench with a flexed lumbar spine while waiting to perform. We recently quantified the loss of compliance in the lumbar spine with bench sitting between bouts of athletic performance (Green, Grenier, and McGill, 2002) in elite volleyball players. Viscosity is also a consideration in prolonged postures because internal friction increases with prolonged static postures. Sitting in this way, and the associated changes in stiffness and

viscosity, are detrimental to athletes' performance and increase their risk of injury. We have been successful in reducing pain in professional basketball players and people in occupational groups by providing lumbar supports and chair seat pan adjustments. We address this issue more completely in chapter 11.

Anatomical Flexible Beam and Truss: Muscle Cocontraction and Spine Stability

The osteoligamentous spine is somewhat of an anatomical paradox: it is a weight-bearing, upright, flexible rod. Observationally, the ability of the joints of the lumbar spine to bend in any direction is accomplished with large amounts of muscle coactivation. Such coactivation patterns are counterproductive to generating the torque necessary to support the applied load. Coactivation is also counterproductive to minimize the load penalty imposed on the spine from muscle contraction. Researchers have postulated ideas to explain muscular coactivation, including that abdominal muscles are involved in the generation of IAP (Davis, 1959) and that they provide support forces to the lumbar spine via the LDF (Gracovetsky, Farfan, and Lamy, 1981). These ideas have not been without opposition (see previous sections).

Another explanation for muscular coactivation is tenable. A ligamentous spine will fail under compressive loading in a buckling mode, at about 20 N (Lucas and Bresler, 1961). In other words, a bare spine is unable to bear compressive load! The spine can be likened to a flexible rod that will buckle under compressive loading. However, if the rod has guy wires connected to it, like the rigging on a ship's mast, although more compression is ultimately experienced by the rod, it is able to bear much more compressive load because it is stiffened and therefore more resistant to buckling. The cocontracting musculature of the lumbar spine (the flexible beam) can perform the role of stabilizing guy wires (the truss) to each lumbar vertebra, bracing it against buckling. Collective work by Crisco and Panjabi (1990); Cholewicki and McGill (1996); Cholewicki, Juluru, and McGill (1999); and Gardner-Morse, Stokes, and Laible (1995) has quantified the influence of muscle architecture and the necessary coactivation on lumbar spine stability. The architecture of many torso muscles is especially suited for the role of

stabilization (Kavcic, Grenier, and McGill, 2004; Macintosh and Bogduk, 1987; McGill and Norman, 1987).